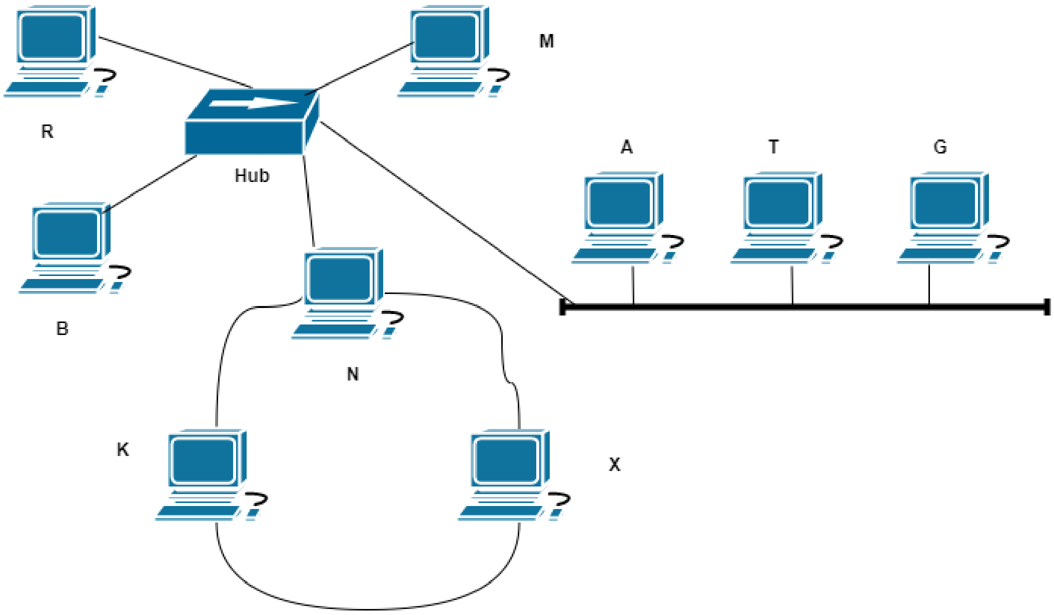
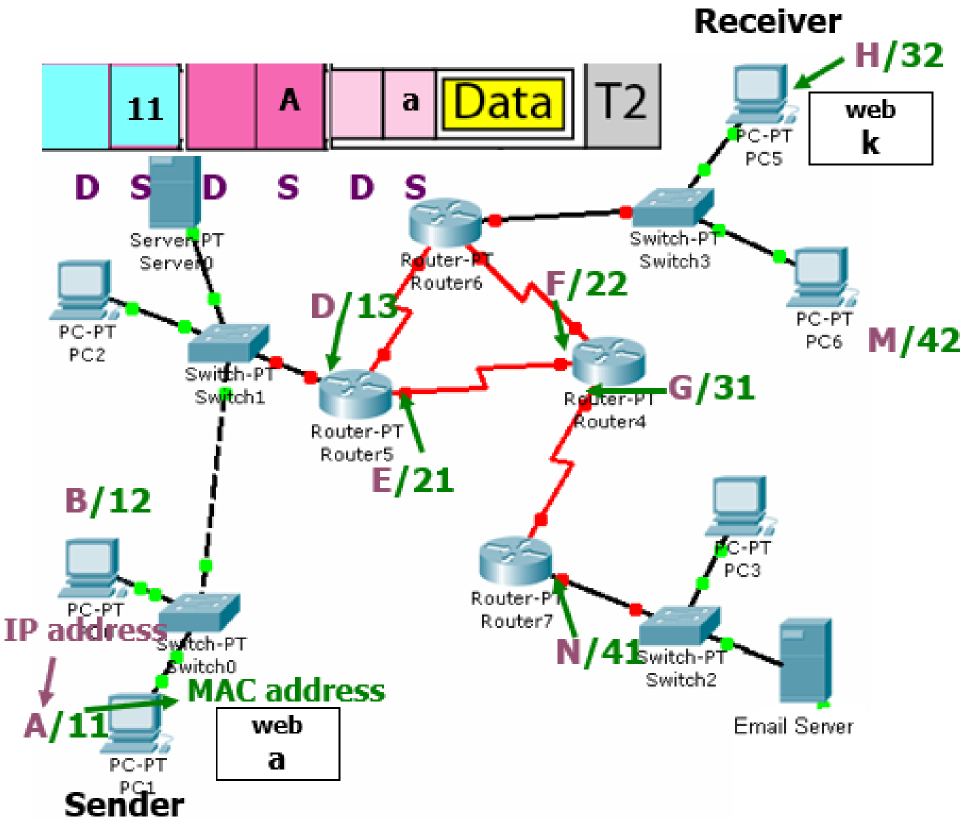
1. In the following figure, you are given a hybrid topology. **[Marks: 4 + 1 = 5]**
   a. Identify all the basic topologies from this hybrid topology and write down all network components (PCs, Hub, Router) associated with each topology.
   [Format: Topology1: A, B, C; Topology-2: D, E, F].
   b. Write down the list of all network components that need to be shutdown to make the whole system down. Here you have to minimize the number of nodes that need to be shutdown to achieve this.



2. You are given a network in the following image. Write down the port, IP, and MAC addresses in the PDUs for communication from the sender to the receiver in the given table.
   [Marks: 6]



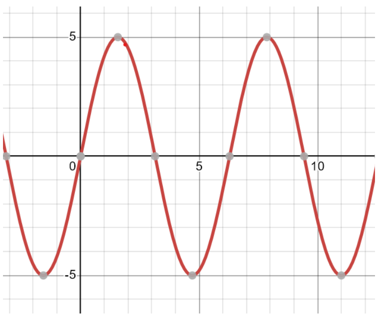
MAC Address		IP Address		Port Address	
Destination	Source	Destination	Source	Destination	Source

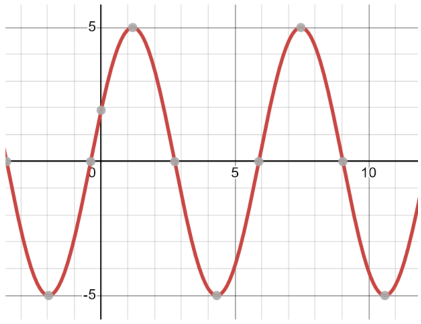
3. Here you are given some addresses. Identify whether the addresses are in the right format. If wrong, identify the error. **[Marks: 4x1 = 4]**

a. IP address: 75.45.256.14
b. MAC address: 50-BE-F6-44-AC-55
c. Server port address: 60000
d. Specific address: https://bux-home.bracu.ac.bd/

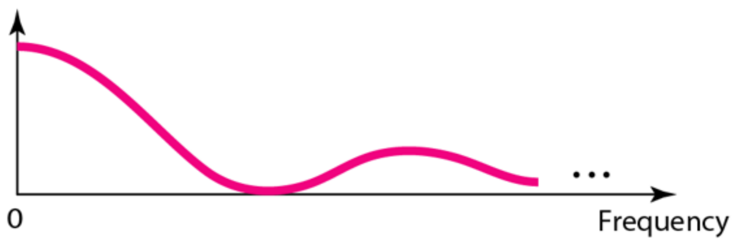
1. For your broadband internet line, you measured the SNR_{dB} to be 5 and the channel bandwidth to be 2 MHz. For the broadband internet line of BracU campus, you measured the SNR_{dB} to be 50 and the channel bandwidth to be 2 MHz. **[Marks: 4 + 2 + 4 = 10]**
    a. Now for both cases find the theoretical channel capacity using the exact formula and the approximate/simplified formula (under high SNR assumption, total 4 results).
    b. Compare the validity of the the simplified formula is two cases assuming high SNR in both cases (that is how close the values are). [Hint: find whether error rate (abs(actual-approx)/actual) < 5 %]
    c. What are the appropriate signal levels in lines and in both the following scenarios (total 4 results) using the results in question 1(a) ?
        i. First, assume the upper limit of capacity gives good performance.
        ii. Second, consider usual choice of capacity that gives better performance.

2. Identify True/False and provide explanations for the answer. **[Marks: 5x1 = 5]**
    a. The two sine waves in the following are in same phase.



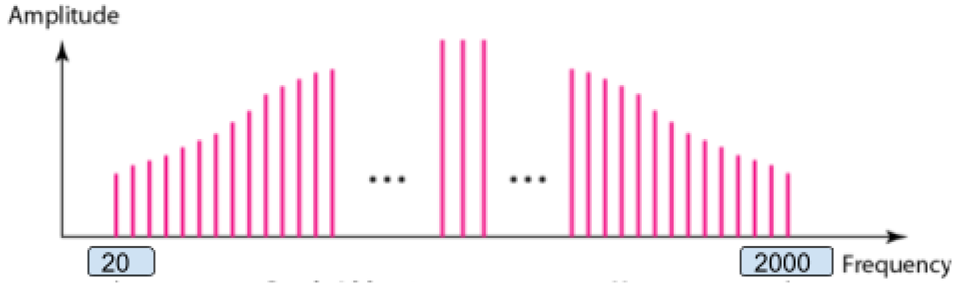


    b. For sending data 6,000 km away at 2.4×10^8 m/s the propagation delay is 30 ms.
    c. This digital signal is periodic.



    d. Ideal SNR is zero which is very common in real life.

e. The bandwidth of the signal is 2000.



1. **For the given data, use**
    a. **Design 4B/5B block coding substitution map, so that there is a maximum of two leading zeros and a maximum of one trailing zeros. [Marks: 7]**
    b. **Draw the signal after substitution with your mapping. [Marks: 2]**
    c. **Apply Manchester and D-Manchester line codings after substitution with the convention followed and provide the plots. [Marks: 3 + 3]**

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